# Jian Feng

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## Skills

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| **Languages:** | Python, Java, C/C++, C#, Go, Rust, TypeScript, SQL, Bash |
| **Frameworks & APIs:** | .NET, Spring Boot, FastAPI, React, Node.js, REST, WebSockets |
| **Data & Cloud:** | PostgreSQL, MySQL, MongoDB, Redis, Azure, AWS, GCP, Docker, Kubernetes |
| **Tools:** | Terraform, GitHub Actions, Linux, Katalon, Selenium, PyTorch, Hugging Face, Power BI, Power Automate, Excel, Figma, Jira/Atlassian, Confluence, Notion |

## Experience

**Manulife Financial Corporation** — Toronto, ON, Canada
*Software Engineer* — *May 2026 – Aug 2026*

- Partnered with Canadian segment and technical teams to refine reusable ServiceNow workflows for **5K+ developers**, automating **1K+ daily operations**; integrated live AKS and certificate checks into a **Foundry LLM agent** to flag GitOps deployment mismatches.
- Replaced plaintext GitHub credentials with **GitHub App token exchange** (PyGithub), adding team-scoped validation, least-privilege permissions, and audit trails.

**University of Waterloo** — Waterloo, ON
*Research Assistant* — *Jan 2026 – Apr 2026*

- Trained four DeBERTa expert models and co-designed an aggregator with an advisor; neutral-context training reduced medical/education false positives, with **sub-50ms inference on a T4**.
- Evaluated the ensemble on benchmark data against fixed-threshold fusion and industry models, reaching **80% F1 and 95%+ end-to-end accuracy**; red-teamed against Llama-3-8B on **5K prompts**.

**Treasury Board Secretariat** — Toronto, ON, Canada
*Software Developer* — *Sep 2025 – Dec 2025*

- Automated duplicate/invalid job-title cleanup for a **5K-firm dataset**, reducing manual processing time **85%**; tuned SQL procedures and indexes to cut report load time **30%**.
- Migrated **30 Selenium tests across five suites** to Katalon on Azure DevOps CI/CD; executive demo drove adoption by **two QA teams**.

**Waterloo Aerial Robotics Group** — Waterloo, ON
*Autonomy Member* — *Jan 2025 – Apr 2025*

- Rebuilt telemetry for the UAV ground-station console (React/TypeScript, Python WebSocket, ROS), restoring **5 dashboard panels** and adding session record/replay for **4.2 hours** of flight logs within a **64 MiB** memory bound.

**Lansi Institute of Artificial Intelligence in Medicine** — Remote
*Data Analyst* — *May 2024 – Dec 2024*

- Parallelized preprocessing of **100GB of multi-omics data** with per-sample Python multiprocessing, cutting runtime **89%, from days to hours** for an international T2D research collaboration.

**Sanyang Technology Co., Ltd.** — Guangdong, China
*Software Engineer* — *Jun 2024 – Nov 2024*

- Translated business-analyst requirements into implementation design and built a Java/Spring job-shop scheduling algorithm, APIs, and interactive Gantt renderer end-to-end.
- Demoed the workflow, secured team sign-off, and tested against production data; exposed reporting through Spring Boot/MySQL APIs for dashboards and downstream ETL.

## Education

**University of Waterloo** — Sep 2023 - Aug 2027
*Bachelor's degree, Computer Science, Finance, and Statistics* — *Waterloo, ON, Canada*

## Projects

**Legos | Rust HFT Trading System**

- Designed a compile-time-generic trading pipeline with interchangeable market feeds, message buses, order books, strategies, risk checks, and execution gateways.
- Implemented low-allocation market-data structures, backtest/testnet adapters, and measured 10M in-process ticks at 70 ns p99 latency, excluding network I/O.

**MiniKV | Go**

- Built an LSM key-value store with a CRC32C write-ahead log, skip-list MemTable, compressed SSTables, sparse indexes, and Bloom filters.
- Added snapshot reads, serializable optimistic concurrency, and asynchronous HTTP/gRPC replication with sequence-based recovery.

**WLP4 Compiler | C++17**

- Built a multi-pass compiler with lexical analysis, LR parsing, semantic/type checking, and ARM64 code generation for control flow, procedures, pointers, and arrays.
- Reduced generated executable size 79.6% versus the reference compiler across 65 benchmark programs; verified correctness with byte-identical regression tests.

## Achievements

- **Velocity Co-op Problem Award** (top submission in cohort); Moonshot Hackathon Winner
- As UX Research Hub Director, owned community growth and programming end-to-end: ran member research, secured a Google UXR guest speaker ( **50 attendees**, **50 new members**), grew membership **32× (10→318)**, and managed event funding with the executive team.